

PRANAV SASANK PALIVELA

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PROFILE SUMMARY

AI/ML Engineer and B.Tech graduate specializing in building end-to-end AI applications powered by Large Language Models, Generative AI, Retrieval-Augmented Generation (RAG), and multi-agent systems. Proficient in Python, LangChain, LangGraph, PyTorch, with full-stack expertise in the MERN Stack. Strong foundation in machine learning, deep learning, NLP, reinforced through Stanford and DeepLearning.AI certifications. Passionate about developing intelligent, production-ready AI solutions and seeking Full-Stack AI Engineer, AI Application Engineer, LLM Engineer, or Generative AI Engineer roles.

EDUCATION

CMR Institute of Technology

Bachelor of Technology - Computer Science (Artificial Intelligence and Machine Learning)

CGPA: 8.54

Nov 2022 – Jul 2026

Narayana Group of Institutions

Intermediate Education (MPC)

Score: 97.4%

Mar 2019 - May 2022

TECHNICAL SKILLS

Programming Languages: Python, C, C++

AI & Machine Learning: PyTorch, LangChain, LangGraph, NumPy, Pandas, Scikit-learn

AI / LLM Concepts: Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Vector Embeddings, Prompt Engineering

Tools & Frameworks: FastAPI, ChromaDB, Git, REST APIs

Web Development: PERN Stack, Redux, Zustand, HTML5, Tailwind CSS, JavaScript (ES6+), REST APIs, PostgreSQL

PROJECTS

AI Personal Agent Workspace | *Python, FastAPI, Next.js, LangChain, ChromaDB, OpenAI API* | [GitHub Link](#)

- Architected a full-stack, production-ready multi-agent AI productivity platform with a **FastAPI** backend and **Next.js 16 / React 19** frontend, featuring **JWT authentication** and per-user data isolation.
- Engineered a **dual RAG pipeline** using OpenAI text-embedding-3-small and **ChromaDB** with two independent vector stores, one for GitHub repo intelligence (auto-clones & indexes any public repo) and one for personal document Q&A supporting PDF and Markdown uploads.
- Optimized API cost and latency via an **in-memory query cache**, fixed-size chunking strategy, and a rolling **5-message context window** for conversation history management.

Bigram Language Model | *Python, PyTorch* | [GitHub Link](#)

- Built a character-level **Bigram Language Model** from scratch using PyTorch to generate unique and plausible names, demonstrating hands-on understanding of **neural network fundamentals** and probabilistic sequence modelling.
- Implemented the full training loop, forward pass, cross-entropy loss, and gradient-based optimization without any pre-trained components.

Hand Gesture Volume Control | *Python, OpenCV, MediaPipe* | [GitHub Link](#)

- Developed a real-time **computer vision** application that controls system volume via hand gestures, leveraging Google's **MediaPipe** for landmark detection and **OpenCV** for video stream processing.
- Mapped the Euclidean distance between the thumb and index finger to the system volume range, enabling **touchless, low-latency gesture-driven control**.

CERTIFICATIONS

The Web Developer Bootcamp 2026 – MERN Stack | Udemy | Jul 2026

Foundation: Introduction to LangChain – Python | LangChain Academy | Feb 2026

Supervised Machine Learning: Regression and Classification | DeepLearning.AI, Stanford University | Sep 2024

Advanced Learning Algorithms | DeepLearning.AI, Stanford University | Nov 2022